

7(a)	1 symbols $\mathbf{B} / \mathbf{X}^{\mathbf{B}}$ = (allele for) black (fur) $\mathbf{G} / \mathbf{X}^{\mathbf{G}}$ = (allele for) ginger (fur) ; parents phenotypes tortoiseshell female ginger male 2 parents genotypes $\mathbf{X}^{\mathbf{B}}\mathbf{X}^{\mathbf{G}}$ $\mathbf{X}^{\mathbf{G}}\mathbf{Y}$; 3 gametes $\mathbf{X}^{\mathbf{B}}$ $\mathbf{X}^{\mathbf{G}}$ $\mathbf{X}^{\mathbf{G}}$ $\mathbf{Y}$; 4 offspring genotypes $\mathbf{X}^{\mathbf{B}}\mathbf{X}^{\mathbf{G}}$ $\mathbf{X}^{\mathbf{B}}\mathbf{Y}$ $\mathbf{X}^{\mathbf{G}}\mathbf{X}^{\mathbf{G}}$ $\mathbf{X}^{\mathbf{G}}\mathbf{Y}$; 5 offspring phenotypes tortoiseshell female black male ginger female ginger male ;	5
7(b)	any three from: 1 TYR codes for <u>tyrosinase</u> ; 2 converts tyrosine into DOPA ; 3 converts DOPA into dopaquinone ; 4 dopaquinone converted to melanin ;	3

Question	Answer	Marks
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